

Received



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

FURTHER MATHEMATICS
PAPER 1

9187/1

NOVEMBER 2014 SESSION

3 hours

Additional materials:

Answer paper

Graph paper

List of Formulae

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

There is no restriction on the number of questions which you may attempt.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

Questions are printed in the order of their mark allocations and candidates are advised to attempt questions sequentially.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 3 printed pages and 1 blank page.

Copyright: Zimbabwe School Examinations Council, N2014.

- 1 Solve the differential equation $x \frac{dy}{dx} + 3y = x^4$ given that $y = 1$ and $x = 1$. [5]

- 2 Show that $\int_0^3 \frac{7x^2 + 11x + 13}{(3x+4)(x^2+9)} dx = \frac{1}{3} \ln 26 + \frac{\pi}{12}$. [10]

- 3 Find the general solution of the differential equation

$$4 \frac{d^2y}{dx^2} + \frac{4dy}{dx} + y = 2 \sin 2x. \quad [10]$$

- 4 (a) Solve the simultaneous equations

$$\begin{aligned} 2x - y - 4z &= 0 \\ 3x + 4y + 12z &= 0 \\ 7x - 9y + 8z &= 0 \end{aligned}$$

- (b) Give the geometrical interpretation of the result in (a). [10]

- 5 (a) If $I_n = \int \sec^n x dx$, show that the reduction formula for

$$I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}, n \geq 2. \quad [7]$$

- (b) Evaluate $\int_0^{\frac{\pi}{4}} \tan^6 \theta d\theta$ exactly using the reduction formula

$$\int \tan^n \theta d\theta = \frac{1}{n-1} \tan^{n-1} \theta - I_{n-2}. \quad [5]$$

- 6 (a) Prove the identity

$$\cosh^2 x + \sinh^2 x = \cosh 2x. \quad [2]$$

- (b) (i) Find the stationary values of the function $y = 25 \cosh x - 7 \sinh x$ and investigate their nature.

- (ii) Hence sketch the graph of the function in (i). [10]

- 7 (a) Show that if x is real, the function $\frac{x^2}{x^2 + x - 6}$ cannot have values between 0 and $\frac{24}{25}$. [4]

- (b) Sketch the graph of the function in (a) showing all asymptotic lines and possible intercepts with the axes. [9]

- 8 The points A, B and C have position vectors $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $7\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}$ and $\mathbf{i} + \mathbf{j} + 3\mathbf{k}$ relative to the origin O.

(a) Find

- (i) the Cartesian equation of the line AB,
- (ii) the vector equation of plane ABC,
- (iii) the area of triangle ABC.

[9]

(b) A point D has position vector $2\mathbf{i} - \mathbf{j} - 6\mathbf{k}$.

- (i) Find the distance of the point, D, from the plane ABC.
- (ii) Hence find the volume of the tetrahedron ABCD.

[5]

- 9 (a) Show that

$$\sum_{r=2}^n \frac{r+3}{r^3-r} = 1\frac{1}{2} - \frac{n+2}{n(n+1)}.$$

[10]

(b) Write down

- (i) the 5th term,
- (ii) the sum of the first 5 terms and,
- (iii) the limit of this sum as $n \rightarrow \infty$ in the series.

[5]

10

- (a) Given that the set $G = \{0; 2; 4\}$ is a group under addition modulo 6, construct the group table and verify that it is commutative.

[4]

- (b) (i) Prove that the set of the matrices of the form $\begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix}$, where r is a positive real number, forms a group under ordinary matrix multiplication.

(ii) Interpret the group in (i) geometrically.

(iii) Show that the group $(\mathbb{R}^+; \times)$, (positive real numbers under multiplication), is isomorphic to this matrix group.

[14]